

Homework #2 — Exploratory Data Analysis (15 pts)

MATH 240 — Introduction to Statistics, Fall 2026

Due Date: Friday, 9/18 at 11:59pm ET

Instructions

- Please submit this assignment to Gradescope as two .pdf files. One will be of the textbook problems, while the other will be of the Palmer Penguins questions. Any submissions not in this format are subjected to point deductions.
- Responses can be hand-written or typed for the textbook problems. You may also use a knitted Quarto file to complete these questions. Parts II and III will be done using RStudio.
- Certain questions will ask you to do the problem in a specific way. Please adhere to these instructions or points will be deducted.
- For all questions involving manual calculations, please show all work and round answers to four decimal places.
- For all questions involving code, please provide your code along with the output.
- Refer to syllabus for late work deductions.

Background

We will now focus on exploratory data analysis, where we will do the following:

- Data manipulation in R
- Analysis of graphs using measures of center and spread
- Creation of graphs and its details using R

Part I: Textbook Problems (7.5 pts)

Complete the following problems:

1. Exercises from Chapter 4: #6, #9
2. Exercises from Chapter 5: #2, #16, #20

Part II: Palmer Penguins (4.5 pts)

We continue our exploration of a dataset consisting of penguins! Specifically, we will be looking at various penguin species observed on three islands in the Palmer Archipelago, Antarctica. Note that we will be using the penguins dataset from the palmerpenguins package.



We will be using R to work with the data. For this question, you will turn in a separate, knitted Quarto document with code and output related to some data manipulation and EDA we will perform. Follow all instructions below.

Create a new Quarto document. Title the document “HW2, Parts II and III” and add your name in the Name box. Use the following code to bring the penguins dataset into your R environment:

- `library(palmerpenguins)`
- `library(dplyr)`
- `library(ggplot2)`
- `data(penguins)`

Again, we will use the penguins dataset for parts II and III.

Tutorial: dplyr Verbs

There is a grammar to data manipulation, which relies on the dplyr package. There are 5 “verbs” that we can use on a given dataset to perform certain actions on the data. Below is a list of the functions and their actions:

- `filter()`: Keeps observations based on some condition
- `arrange()`: Arranges dataset based on order of a chosen variable
- `select()`: Selects certain columns from dataset
- `mutate()`: Creates new variable in dataset based on other variables
- `group_by() |> summarize()`: Groups data by chosen categorical variable and summarizes based on certain metric

When using these functions, we will pipe a dataset into the function we are using. For instance, if we want to order the dataset based on the bill length, we can write the following code:

- `penguins |> arrange(bill_length_mm)`

Note that we do not need to specify penguins when referring to the bill length. The `arrange` function, as well as the other dplyr functions, bring in the penguins dataset first (which is what the `|>` operator does), then it knows to find the specific column in that dataset.

More information about the actions can be found [here](#). Additionally, you can put `?dplyr::_____` in the console, where `_____` is the function name you want to investigate, to learn more about the function and its arguments. We will practice using these functions below:

1. (1 pt) Use the `filter()` function to output a dataset that only contains penguins from Torgersen island.
 - Hint: `==` will be a useful operation within the `filter()` function to compare a variable to a phrase you want to match.
2. (0.75 pts) Use the `arrange()` function to order the dataset by the penguin's body mass in decreasing order.
3. (0.75 pts) Use the `select()` function to output a dataset with only its first three columns.
4. (1 pt) Use the `mutate()` function to create a new column that labels a penguin as "long-billed" if their bill length is longer than or equal to 45 mm, and "short-billed" otherwise.
 - Hint: Use the `ifelse(__, __, __)` function within `mutate`, where the three respective arguments are:

- Condition: `variable >= count`
- Output if condition is satisfied: label
- Output if condition is not satisfied: label

5. (1 pt) I can use the `group_by()` |> `summarize()` code as follows:

- `group_by(variable) |> summarize(count = n())`

The variable I want to group by goes in the `group_by` function and the metric I want to use can be formatted as the name (of my choice) equaling a function with a blank set of parentheses. `n()` gives counts, but I can also use `mean(numeric variable)`, `summary(numeric variable)`, and other functions to operate on data in those groups. To that end, find the means of flipper lengths across penguin species.

Part III: Palmer Penguins (3 pts)

We continue where we left off with regarding the Palmer Penguins in Part II. This time, we will create some plots.

A plot is created by combining two functions together at the bare minimum:

- `ggplot(data = ___, mapping = aes(x = ___, y = ___, color = ___, ...))`
- `geom_bar()`, `geom_boxplot()`, `geom_histogram()`, `geom_scatter()`

Note that in the `ggplot()` function, we need to specify the data frame we are using, and which variables we are using to match some aesthetic in the plot. If one variable is used, we just need to use `x`. If two are used, we use `x` and `y`. If we want to change the fill, color, shape, etc. with a variable from the dataset, we can specify that here too.

For one of the plotting functions, we can add to the `ggplot()` function. For example, if we want a scatter plot of the bill length and bill depth, we can use the following code:

- `ggplot(penguins, aes(x = bill_length_mm, y = bill_depth_mm)) +
 geom_point()`

Answer the following questions below:

1. (1 pt) Create a histogram of the flipper length, specifying the number of bins to be 20 (e.g. `bins` is an argument in the `geom_histogram()` function). Then, describe the center, spread, and skewness of the distribution of flipper length in 2-3 sentences.

2. (1 pt) Create a scatter plot of the bill length and body mass. Identify the strength and direction of association.
3. (1 pt) Create a side-by-side box plot of bill lengths across the different islands. Briefly compare and contrast the centers and spreads of the bill lengths across islands in 1-2 sentences.

As extra practice (not to be graded), try to add labels, titles, themes, etc. to the ggplots. You will find the `labs()` and `theme()` functions to be useful here.